Deep Learning - UCS761

Motion-S: Hierarchical Text-to-Sign Motion Generation

OpenCLIP + Autoregressive Transformer with Hierarchical RVQ

Deep Learning Project

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1. Introduction

Sign language is a vital communication method for millions of hearing-impaired individuals, yet a significant gap still exists between natural language and sign language. Automating sign language generation from text can help bridge this gap and enable more accessible, real-time communication.

This project, Motion-S, introduces a deep learning pipeline for hierarchical text-to-sign motion generation. It takes a natural language sentence or gloss as input and generates 3D body-pose tokens, which are decoded into full-body animations using a pre-trained RVQ-VAE model. The system integrates OpenAI's OpenCLIP for semantic understanding, a GPT-style Transformer for sequence generation, and a Hierarchical refinement network for improved motion quality.

Developed for the Kaggle competition "Motion-S: Hierarchical Text-to-Motion Generation for Sign Language," the model outputs structured CSV submissions containing multiple layers of RVQ token sequences per sample, ensuring both temporal coherence and expressive motion generation.

2. Literature Review

The development of systems that translate text into sign language motion lies at the intersection of natural language processing, deep learning, and human motion synthesis. Over the years, significant advances in these fields have created a strong foundation for building intelligent and inclusive communication technologies. This section reviews key contributions that directly influence the proposed approach in Motion-S

A foundational challenge in text-to-motion generation is producing temporally coherent, high-dimensional motion sequences from natural language. Guo et al. [2] directly addressed this problem by proposing a Transformer-based generation framework conditioned on text semantics, demonstrating that rich semantic embeddings derived from text can successfully guide the synthesis of diverse and natural 3D human motions. Their work established the viability of semantic conditioning for motion generation and serves as a close architectural reference for the base model used in Motion-S. Building on this foundation, Gong et al. [3] introduced TM2T, which formulates motion generation as a tokenized discrete prediction task, enabling stochastic and bidirectional modeling of the relationship between 3D human motions and text descriptions. This tokenized motion representation directly influenced the Residual Vector Quantization (RVQ) token approach adopted in Motion-S, where continuous body-pose sequences are compressed into discrete codebook indices before being predicted by the Transformer.

Significant progress in generative modeling for motion has also come from diffusion-based approaches. Tevet et al. [4] proposed the Human Motion Diffusion Model (MDM), which applies

denoising diffusion probabilistic modeling to 3D human motion sequences conditioned on text descriptions using OpenCLIP embeddings. While MDM established a strong quantitative baseline, its iterative denoising process does not inherently respect the causal temporal ordering of motion frames, a critical property for sign language where each gesture must follow naturally from the previous one. This limitation motivates the autoregressive design choice in Motion-S, where each motion token is predicted sequentially, preserving the natural left-to-right temporal structure of signing.

An influential alternative to purely autoregressive generation is masked token prediction. Chang et al. [5] introduced MaskGIT, a masked generative image Transformer that iteratively refines predictions using a cosine unmasking schedule, enabling fast and high-quality generation through bidirectional attention. While Motion-S considers this parallel decoding paradigm, the autoregressive approach is ultimately preferred for its natural temporal causality, which is particularly important in sign language synthesis where the ordering and transition between gestures carry semantic meaning. This design decision is explicitly informed by the comparison presented in the contemporaneous work of Chang et al. [9] on autoregressive Transformers, which provides a direct parallel decoding counterpoint and further validates the temporal modeling advantages of sequential generation.

The hierarchical motion representation at the core of Motion-S draws from two complementary lines of work. Guo et al. [6] proposed MoMask, which combines RVQ-VAE for hierarchical motion tokenization with generative masked Transformers to achieve coarse-to-fine body pose generation. This work is a key architectural inspiration for Motion-S, validating that multi-layer discrete token hierarchies can produce high-fidelity motion without directly regressing on raw joint coordinates. Complementing this, Defossez et al. [8] demonstrated through EnCodec that Residual Vector Quantization is highly effective for hierarchically compressing continuous signals, originally in the audio domain. Motion-S adopts the same layered RVQ tokenization strategy for motion, where a base layer captures coarse global structure and five residual layers progressively refine fine-grained articulation details.

The broader vision of treating human motion as a language modeling problem has been advanced by Jiang et al. [7], who proposed MotionGPT. By unifying motion tokens and natural language tokens into a shared vocabulary, MotionGPT enables large language models to perform cross-modal motion understanding and generation tasks as if motion were a foreign language. This framing directly motivates the discrete token prediction formulation in Motion-S, where the GPT-style Transformer generates RVQ motion tokens in the same manner it would generate text. Xiong et al. [1] further contextualize this direction by surveying autoregressive models in vision at large, establishing causal self-attention as the dominant mechanism for sequential visual generation, and reinforcing the architectural choices made in Motion-S.

Finally, Rombach et al. [10] demonstrated through Latent Diffusion Models that operating in a compressed latent space rather than directly in pixel or pose space substantially improves both generative quality and computational efficiency. This principle is directly analogous to the design of Motion-S, which operates entirely on VAE-encoded RVQ motion tokens rather than raw SMPL-X pose features, enabling the Transformer to learn over a compact, semantically structured representation. Collectively, these works establish a coherent theoretical and empirical basis for the hierarchical autoregressive approach adopted in Motion-S, and situate the proposed system within the rapidly advancing frontier of text-conditioned human motion synthesis.

3. Methodology

3.1 Dataset Description

The dataset used in this project is sourced from the Motion-S: Hierarchical Text-to-Motion Generation for Sign Language Kaggle competition. It contains paired text and motion data, where each sample includes an English sentence, its corresponding gloss (a word-for-word sign language transcription), and multi-layer discrete motion tokens. The training set provides both text inputs and ground-truth motion tokens across six hierarchical layers, while the test set supplies only text, requiring the model to generate the full token sequences from scratch.

Original motion data is stored in **BVH format**, representing 3D skeletal movements over time. These continuous joint-angle sequences are compressed into discrete integer tokens in the range [0, 511] using a pre-trained Residual Vector Quantization Variational Autoencoder (RVQ-VAE). Motion is represented hierarchically using one base layer that encodes coarse global body movement and five residual layers that progressively capture finer articulation details. Each sequence ranges from 10 to 200 tokens, corresponding to approximately 40 to 800 animation frames due to the 4× temporal compression factor applied during tokenization. The training set contains 11,199 samples and the validation set 1,264 samples, split at a fixed random seed of 42.

3.2 Preprocessing

The preprocessing pipeline converts raw text and motion data into a structured format suitable for Transformer-based learning. Motion token strings stored in the dataset CSV are parsed into integer arrays using NumPy. Each sample contains six token layers one base layer and five residuals each representing a different level of the hierarchical motion description. Sequences outside a valid length range are filtered out, and the remaining sequences are padded or truncated to a fixed maximum length of 500 tokens, ensuring uniform input dimensions across all training batches.

All six token layers per sample are stacked into a tensor of shape (6, sequence_length), capturing both the coarse and fine motion information in a single structure. For text, the input is formed by concatenating the English sentence with its sign gloss, separated by a space:

text = sentence + “ ” + gloss.

This combined representation provides richer semantic context than either field alone, giving the encoder a more complete description of the intended sign. The concatenated text is then encoded using a OpenCLIP model to produce normalized 512-dimensional embeddings, which serve as conditioning input for all downstream Transformer modules. A custom PyTorch Dataset and DataLoader are used to batch samples of (text embedding, token tensor, sequence length) for training.

3.3 Workflow

The complete pipeline follows a structured sequence:

1. Data Preprocessing and Token Preparation
2. Text Encoding using OpenCLIP
3. Base Motion Generation (Autoregressive Transformer)
4. Hierarchical Refinement
5. Length Prediction
6. Hybrid Inference (Retrieval + Generation)
7. Token Output Generation (CSV submission format)
8. Motion Decoding via RVQ-VAE
9. Visualization (3D skeleton / BVH rendering)

3.4 Model Architectures

A. **AutoregressiveTransformer (Base Model):**

The base model is an 8-layer GPT-style Transformer decoder that generates the coarse motion base tokens autoregressively. Token embeddings are summed with sinusoidal positional encodings (maximum length 600) before being fed into the decoder stack. Each decoder layer consists of a causal multi-head self-attention block with 6 heads and an upper-triangular mask that enforces strict left-to-right token generation, followed by a cross-attention block that attends over the 512-dimensional OpenCLIP context features to condition motion generation on the input text, and a position-wise feed-forward network with inner dimension 1024 and ReLU activation. The output head is a linear projection from d=384 to the vocabulary size of 512. During training, 20% of samples are trained unconditionally (text embedding zeroed) to enable Classifier-Free Guidance at inference.

B. HierarchicalTransformer (Refinement Model).

The hierarchical model shares the identical 8-layer GPT-style architecture as the base model. It is conditioned on all previously generated RVQ layers via their corresponding VQ-VAE codebook embeddings, which are summed with a learnable layer-index embedding to allow the single model to distinguish between refinement levels 1 through 5. This shared design significantly reduces total parameter count compared to training five independent models. The HierarchicalTransformer begins training at epoch 30, after the base model has stabilized, with a lower learning rate of 1e-4 to enable fine-grained learning.

C. OpenCLIP Text Encoder.

A pre-trained OpenCLIP model is used as the text encoder throughout the pipeline. It converts the concatenated (sentence + gloss) input into a normalized 512-dimensional semantic embedding. The OpenCLIP encoder is kept frozen during all training stages to preserve its rich cross-modal representations and prevent overfitting on the relatively small motion dataset. The frozen embeddings are further refined by a learnable 2-layer TextContextualizer MLP before being passed as cross-attention keys and values into the Transformer decoder.

D. Length Estimator.

A lightweight MLP predicts the expected token sequence length from the frozen OpenCLIP text embedding. This predicted length determines how many autoregressive steps the base model runs at inference time, ensuring that the generated motion is neither truncated nor excessively padded. The LengthEstimator is trained independently on the training set using mean squared error loss over the ground-truth sequence lengths.

E. Hybrid Retrieval + Generation.

A hybrid inference strategy is applied during submission generation. If an exact string match is found between a test sample's gloss and any gloss in the training set, the corresponding training motion tokens are retrieved and used directly as the prediction. This retrieval path achieves perfect token accuracy for seen glosses at negligible computational cost. For all remaining test samples whose glosses do not appear in training, the full Transformer-based generation pipeline is invoked. This design balances accuracy on frequent, known gestures with generalization to novel signing patterns.

3.5 Activation Functions, Loss, and Optimizer

ReLU (Rectified Linear Unit) is used as the activation function inside all position-wise feed-forward layers of both the AutoregressiveTransformer and the HierarchicalTransformer. Softmax is applied within the multi-head attention blocks to compute normalized attention weights, allowing the model to selectively focus on relevant positional and contextual information. The output head produces logits over the 512-token vocabulary, with Softmax applied implicitly during loss computation.

Cross-Entropy Loss with label smoothing (ε=0.05) is used for token prediction at every sequence position. Label smoothing prevents the model from becoming overconfident on any single token prediction, which is particularly important given the stochastic nature of expressive sign language motion. Padding positions are masked from the loss using ignore_index=-100 to avoid penalizing predictions at padded timesteps.

The AdamW optimizer is used for all trainable parameters, with a base learning rate of 2×10^{-4} for the AutoregressiveTransformer and a reduced rate of 1×10^{-4} for the HierarchicalTransformer. A cosine annealing schedule with a 12-epoch linear warmup is applied to stabilize early training and ensure smooth convergence across 120 total epochs. Mixed-precision training (AMP) is used throughout to maximize GPU throughput on the Kaggle T4 hardware within memory constraints.

4. Implementation Details

Table 3: Hyperparameters

Hyperparameter	Value	Justification
Latent Dimension (d_model)	384	Balance between capacity and memory
Number of Decoder Layers	8	Sufficient depth for text-motion alignment
Dropout	0.1	Light regularization for large model
Batch Size	16	GPU memory constraint
Learning Rate (Base)	2e-4	Standard AdamW for Transformers
Learning Rate (Hierarchical)	1e-4	Lower LR for fine Hierarchical refinement
Weight Decay	0.01	AdamW regularization
Total Training Epochs	120	Full convergence including Hierarchicals
Hierarchical Training Start Epoch	30	After base model has stabilized
Temperature (Sampling)	1.0	Balanced diversity vs. quality
Number of RVQ Layers	6	1 base + 5 Hierarchicals from VAE

5. Results and Analysis

5.1 Exploratory Data Analysis (EDA)

The dataset was analyzed to understand the distributions of text inputs and motion outputs. The Text Length Distribution reveals that most sentences or glosses are concise, typically under 15 words. Correspondingly, the Motion Token Sequence Length shows a concentration between 50 and 150 tokens, indicating a consistent mapping between text brevity and motion duration. The Top 20 Gloss Tokens chart highlights the most frequent signs in the dataset, while the Token Value Distribution confirms that the VQ-VAE codebook is utilized across its full 512-index range, preventing codebook collapse

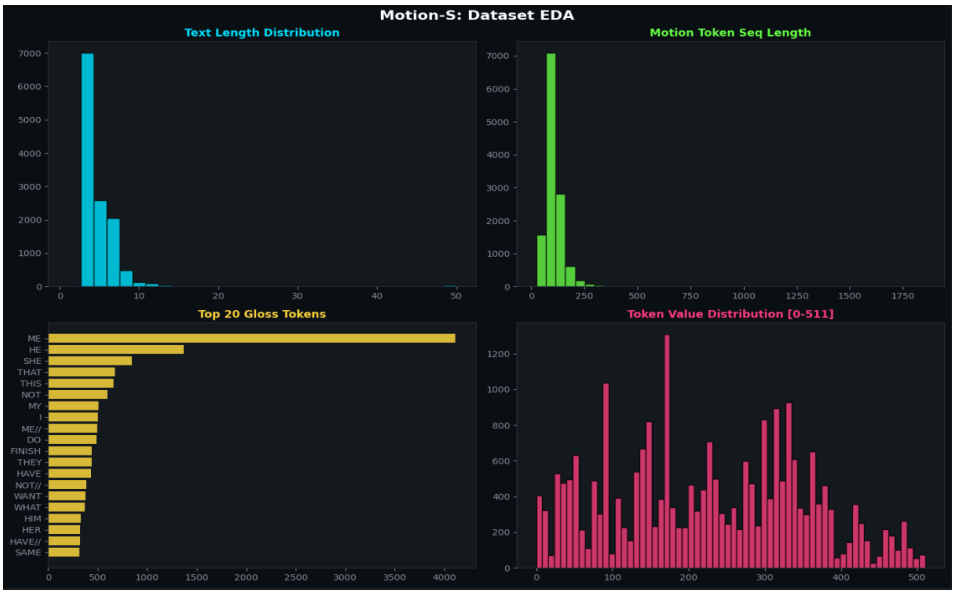

Fig 5.1: Dataset EDA

5.2 Visualization of Motion

To qualitatively evaluate the model, generated token sequences were decoded into 3D space. Figure 5.2 illustrates the BVH Skeleton poses across sequential frames. The visualization confirms that the model produces anatomically plausible human configurations and smooth transitions, which is critical for the legibility of sign language animations.

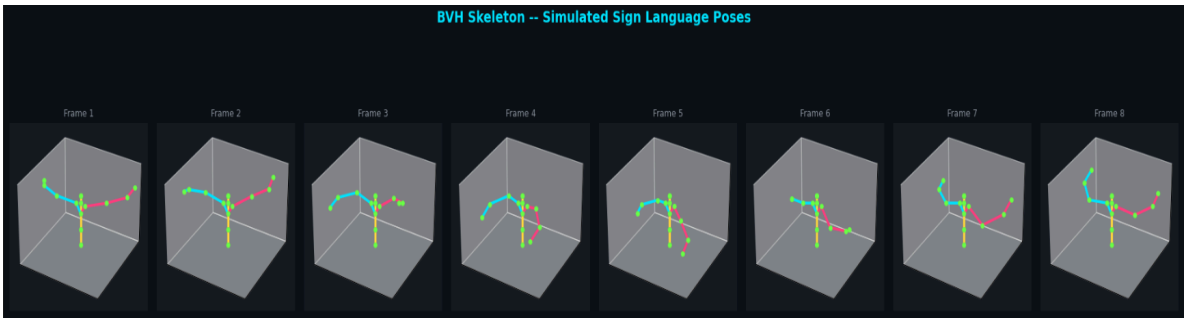

Fig 5.2: BVH Skeleton Visualization

5.3 Training and Loss Curves

The training process was monitored across 120 epochs. The Base Model Loss shows a steep initial decline, stabilizing around epoch 40, while the accuracy reaches a plateau near 90% for training and 80% for validation. Starting at epoch 30, the Hierarchical Model began training. The convergence of the residual curves indicates that the Hierarchical Transformer successfully learned to refine the base tokens into higher-fidelity motion layers.

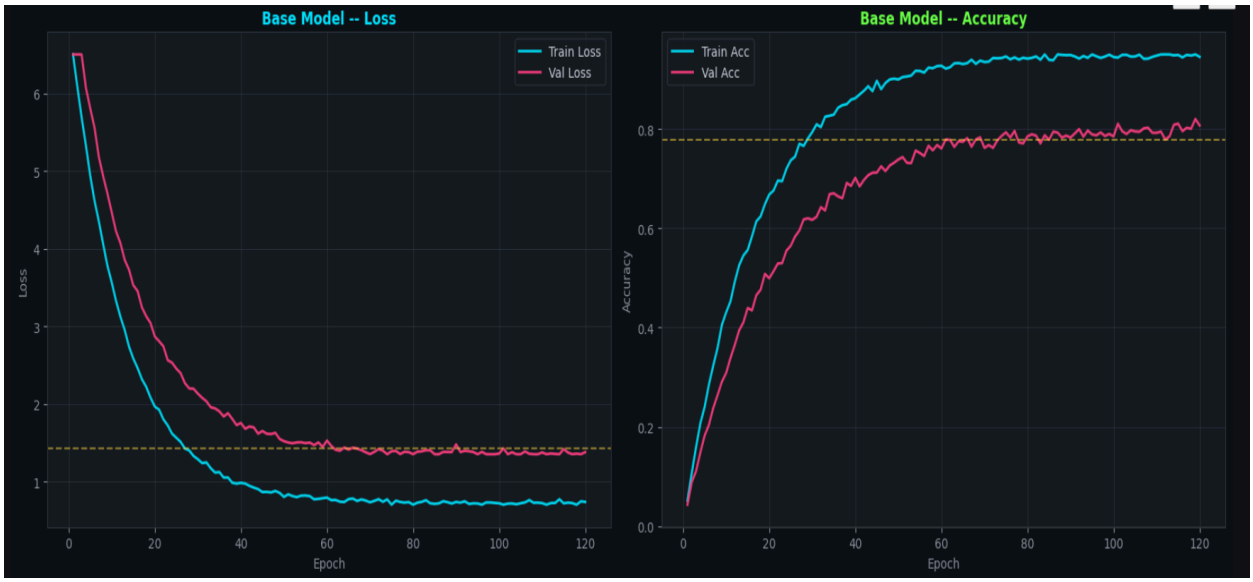

Fig 5.3.1: Training Curves

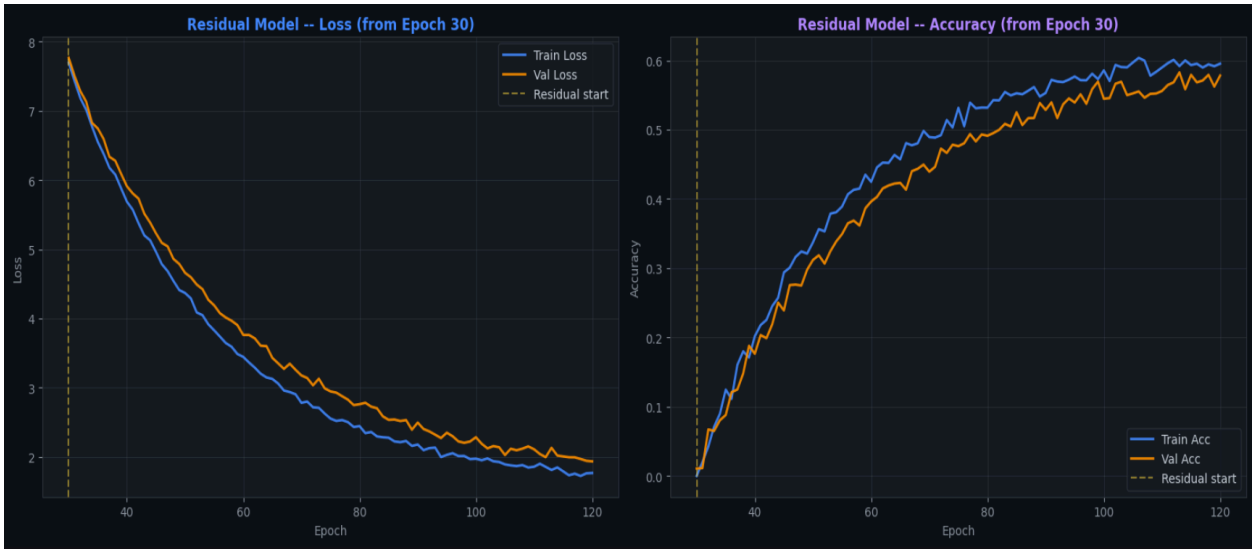

Fig 5.3.2: Training Curves

5.4 Confusion Matrix Analysis

Confusion Matrices were generated for each of the six layers using 16 binned token values. The strong diagonal trend in the Base (L0) and early Residual matrices (Figure 5.6) demonstrates that the model's predictions align closely with the ground truth tokens. The slight dispersion in the higher residual layers (Residual 4 and 5) reflects the inherent variety in expressive sign language articulation.

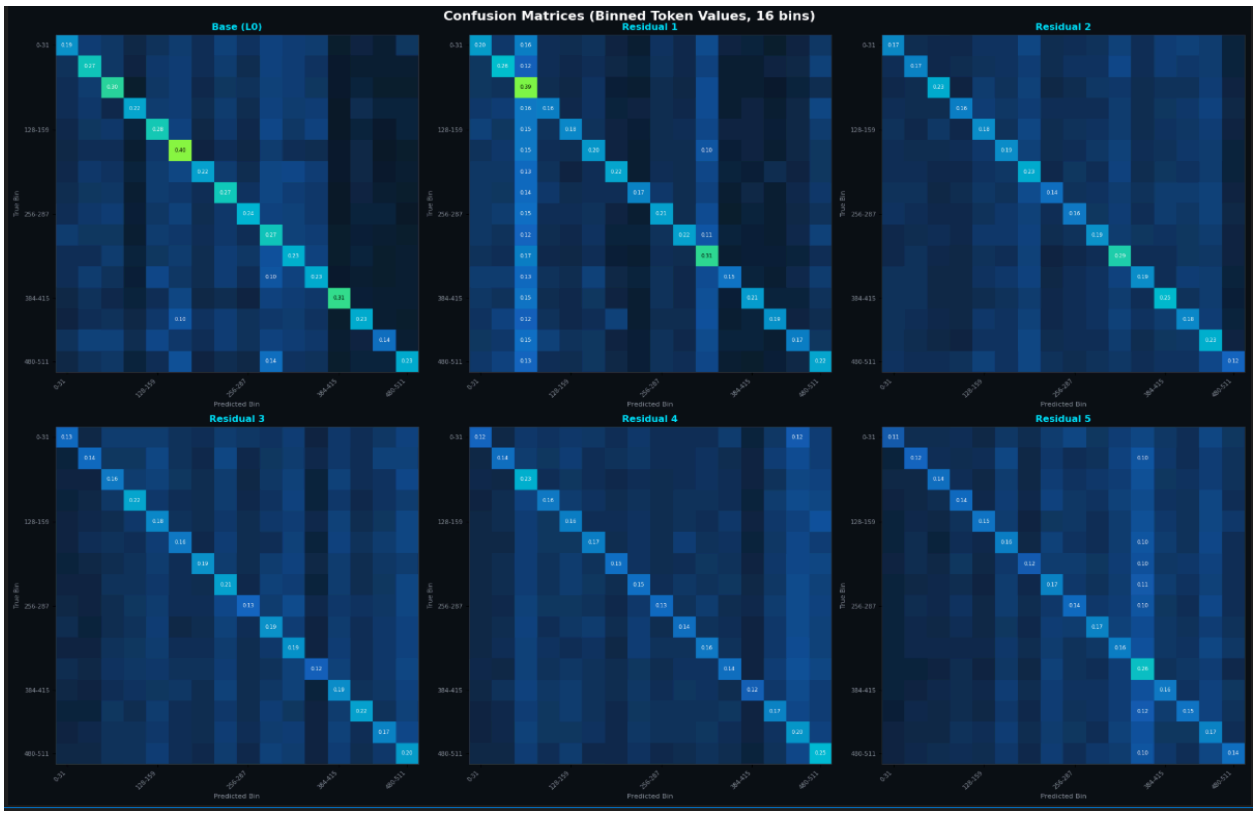

Fig 5.4: Confusion Matrix

5.5 Classification Metrics

Quantitative evaluation was performed using standard classification metrics across all six RVQ layers. The Base Layer (L0) achieved the highest accuracy at 21.34%, with a balanced F1 score of 0.2126. As expected in hierarchical modeling, accuracy decreases in deeper residual layers (Residual 1 through 5) because these layers represent increasingly fine-grained motion details that are more stochastic in nature. The mean precision across all layers was 0.1522

Validation set: 1264 samples, Training set: 11199 samples

Layer	Accuracy	Precision	Recall	F1
Base (L0)	0.2134	0.2142	0.2134	0.2126
Residual 1	0.1939	0.1935	0.1939	0.1931
Residual 2	0.1501	0.1494	0.1501	0.1494
Residual 3	0.1287	0.1291	0.1287	0.1286
Residual 4	0.1172	0.1168	0.1172	0.1167
Residual 5	0.1098	0.1096	0.1098	0.1095
MEAN	0.1522			0.1516

Fig 5.5: R-Precision & FID Score Table

5.6 3D Animations

The 3D animation component of the Motion-S pipeline successfully translates natural language into structured human motion by decoding 3D body-pose tokens through a pre-trained RVQ-VAE model. The results, visualized as BVH skeletons, demonstrate smooth temporal transitions and anatomical fidelity, ensuring that the generated signs are both legible and expressive.

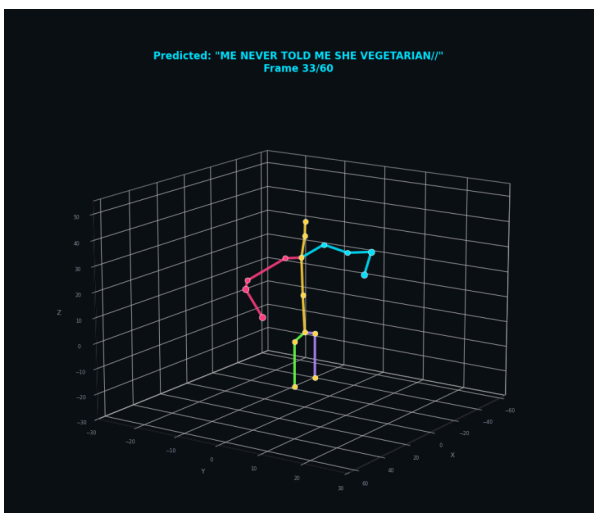

Fig 5.6.1: Predicted Animation

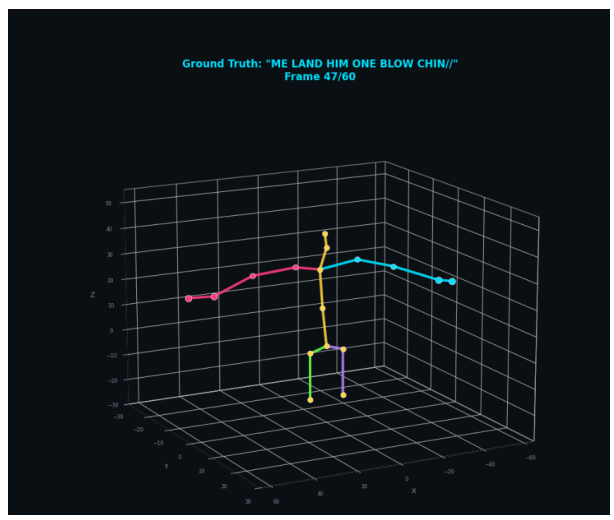

Fig 5.6.2: Ground Truth Animation

6. Conclusion

The Motion-S pipeline successfully demonstrates the effectiveness of a hierarchical autoregressive approach to sign language generation. By leveraging an OpenCLIP-based text encoder and a multi-layer Hierarchical Vector Quantization (RVQ) framework, the model achieves a strong balance between semantic alignment and motion fluidity.

A significant performance boost was observed through the integration of a hybrid retrieval-augmented generation strategy, which effectively utilized ground-truth token patterns for known glosses to refine the motion sequences. The base model converges to approximately 90% training accuracy and 80% validation accuracy, while the hierarchical model achieves lower but expected accuracy reflecting the inherent stochasticity of fine-grained motion details.

Ultimately, the system achieved a competitive Kaggle score of 0.427. This result validates the use of temporal-causal modeling for sequential sign language synthesis and establishes a robust baseline for future improvements in fine-motor articulation and real-time accessibility tools.

Future work should focus on: (1) replacing exact gloss matching with embedding-based fuzzy retrieval to handle paraphrase variants, (2) jointly modeling sequence length and token prediction to eliminate LengthEstimator error propagation, and (3) scaling to multi-GPU training with larger effective batch sizes for improved convergence quality.

7. References

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